Remotely controlled exploration rover with robotic arm

Problem: Challenges in performing hazardous or risk prone activities remotely on earth or under water resulting into loss of human lives. Also, inability to perform extraterrestrial exploration on celestial bodies not feasible to send human.

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Abstract

My problem is that humans may have difficulty in reaching faraway places. For example: People have not yet been able to reach Mars, or people have not been able to dive to the bottom of the Mariana Trench in diving suits, or it is extremely risky to send miners into unknown mines. The rover is a vehicle that will help explore unknown and sometimes even hazardous places (For example: A mine with unstable soil, dangerous gases, or a place with fire). Hence, it will avoid the risk of human life. I started the design of the chassis with 6 wheels. Also, I made sure that all the weight on the arm was at the base and not one servo at each joint. After designing the chassis and arm, I started building. When I finished, I realized that I need to remove the 2 non-motorized middle wheels to reduce drag. Then comes the camera. At first, I thought of using the Raspberry Pi to stream. But then it was found that it has no graphics card and cannot handle the video streaming along with 7 motor controls. So, I used the Jetson Nano and it works better than a computer with a shared graphics card (for steady video streaming) and is almost as good as a computer with a dedicated graphics card. While my solution does work for the exploration purpose, however the vehicle design would require adjustments depending on the environment it is deployed to.

Background Research

How many miners die due to accidents in the mines every year?
Over 15, 000 miners die every year due to accidents or injuries while at work.

How many firefighters die each year due to fire or non - fire emergencies?
About 70 firefighters have died due to fires and other non - fire emergencies over the past two years.

What risks do astronauts face while taking an EVA?
Although no astronauts have died on EVAs, they are still in increasing risk of getting hit by space debris as there is a thick ring of it around the Earth and the number of debris is increasing as more and more rockets go into space.

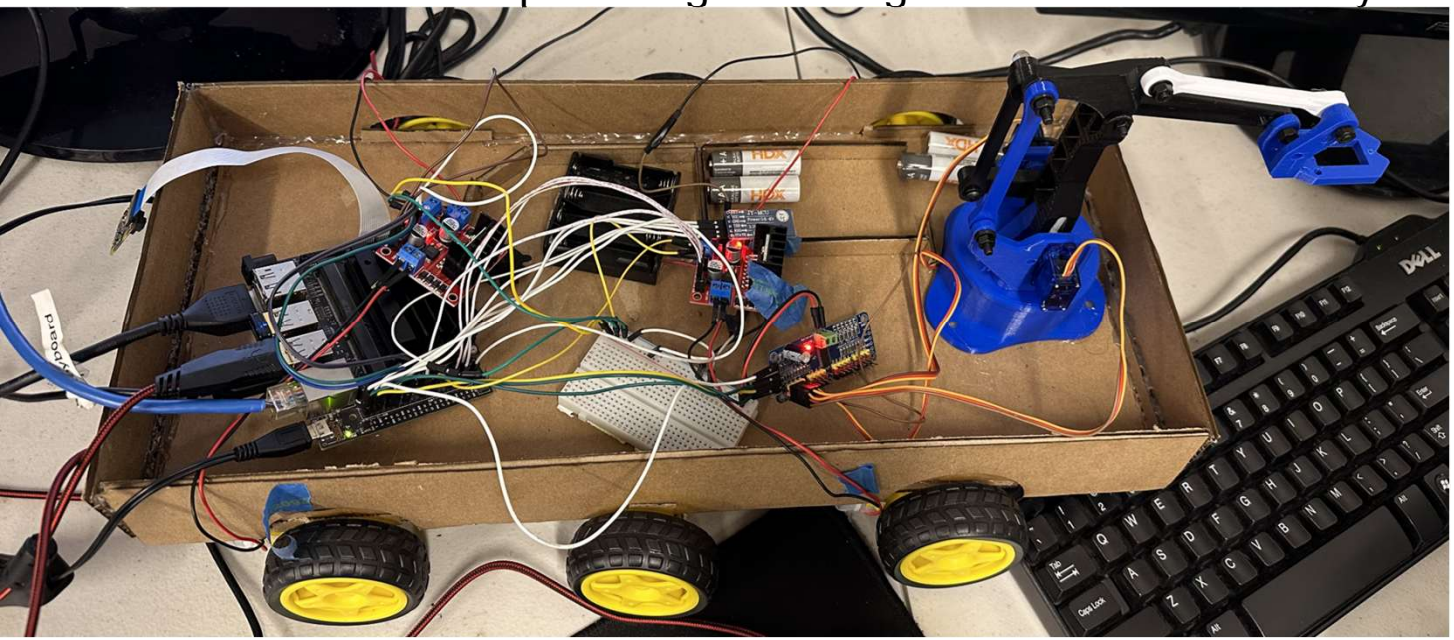
How many divers have died while trying do collect samples for marine research?
About 200 people die while diving all over the world. If they are looking for coral reef samples, new animal species, etc, the robotic arm can do it for them while they control it remotely from above the water and they can avoid putting their lives at risk.

What inspired me to do the project?
My inspiration for the project was all the NASA rovers that have landed on Mars. They have helped me learn about that distant extra-terrestrial body. So now, I want to make a rover that can help explore. With modifications, it can also go to space or even go underwater.

What allows the best live streaming?
A dedicated graphics card and good hardware allows the best live streaming of video. That is because the dedicated graphics card can do all the video streaming work pretty fast and that is all it needs to do at that time. So, it can outperform any shared graphics card.

The Engineering Solution, Prototype or Model to be tested.

It is an exploring robot that has been designed for exploring purposes on land only for the time being. The prototype model is basically a rover that can be controlled remotely with a joystick module. The connection between the control center or joysticks and the rover is a Bluetooth with 98 feet range. The arm is controlled by a total of 2 joysticks and can go left-right, up-down, and forward-back (i.e. 3 degrees of motion). The chassis has a total of 6 wheels and 4 motors to help the rover move and the middle two wheels are for support of better weight distribution. Finally, the camera is the part that will send live feed to control center and will help to navigate through the unknown and maybe dangerous place. (Initial design)



This prototype was built based on the initial design on 29-Aug (please refer to the Log Book entry). The arm was 3D printed with improved design version to move all the servo motors to the base. By this time, the decision on board selection (as Jetson) done.

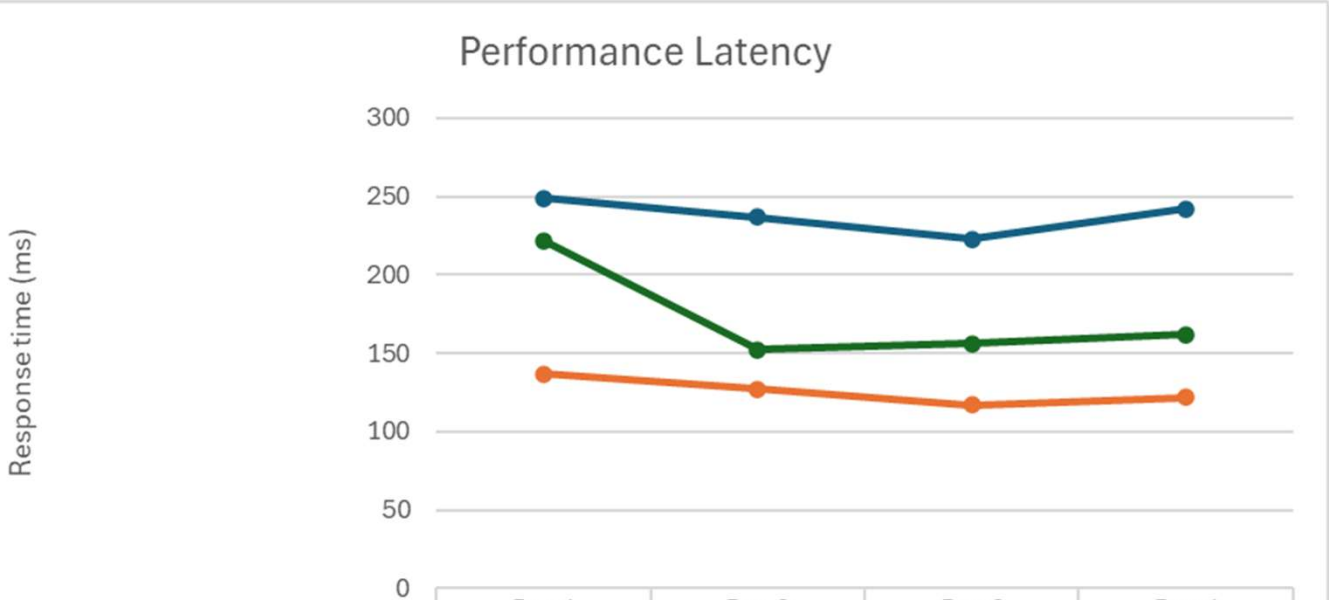
Materials

- Cardboard
- Laser Cutter
- CorelDraw
- 3D printer filament (Generic PLA)
- Prusa i3 MK3 (3D printer)
- 4 Wheels
- 4 DC motors (Adafruit LLC-3777)
- 1 Arduino UNO R3
- Arduino IDE
- 3 Servo Motors
- 1 NVIDIA Jetson Nano (has a dedicated graphics card and a SD card)
- Python 3.8
- Python 3.6
- 1 Raspberry PI camera
- 2 Bluetooth modules (HC05)
- M-M, M-F, and F-F wires
- 2 breadboards
- 3 Parallax 2 axis joystick modules
- Hot Glue
- M4 and M3 screws, washers, and nuts
- 3 Servo Motor horns
- 3 LEDs (to demonstrate status of various stages during boot)
- 3 Resistors
- 1 Voltage regulator Board
- 2 L298N motor driver boards
- 1 Adafruit Servo controller board

Procedure

1. I needed a vehicle with moving arm for exploration and sample collection.
2. So, I designed a base/chassis to support the mass. It had 6 wheels and but then it ended having too much drag and then I had to reduce it to 4 wheels.
3. I researched and tested options between Raspberry Pi and Jetson Nano for the vehicle module. However, test data showed Raspberry Pi could not handle 7 motor control commands along with video streaming. Hence, I decided to proceed with Jetson Nano.
4. After about 8 weeks of work. I had it working for the first time and the camera module was yet to be introduced. By this time, the vehicle control module (python code on version 3.8) was fully developed and tested.
5. I added the camera. The camera streaming code was developed on the python3.6 (with corresponding OpenCV library). While Jetson is running on Python 3.8, however it has backward compatibility with 3.6.
6. Initially I thought of going with RFCOMM (Bluetooth USB chip) based communication between control module (HC05) and Rover Module (RFCOMM). However, Jetson failed to load the required kernel and hence I used with the Bluetooth module for both ends.
7. As my design revisions were done, I tested all the expected functionalities and fine tuned some of the code issues.

Results – Data/Observations



	Run 1	Run 2	Run 3	Run 4
Computer w/o dedicated card	249	237	223	242
Computer w/ dedicated card	137	127	117	122
Jetson Nano	222	152	156	162

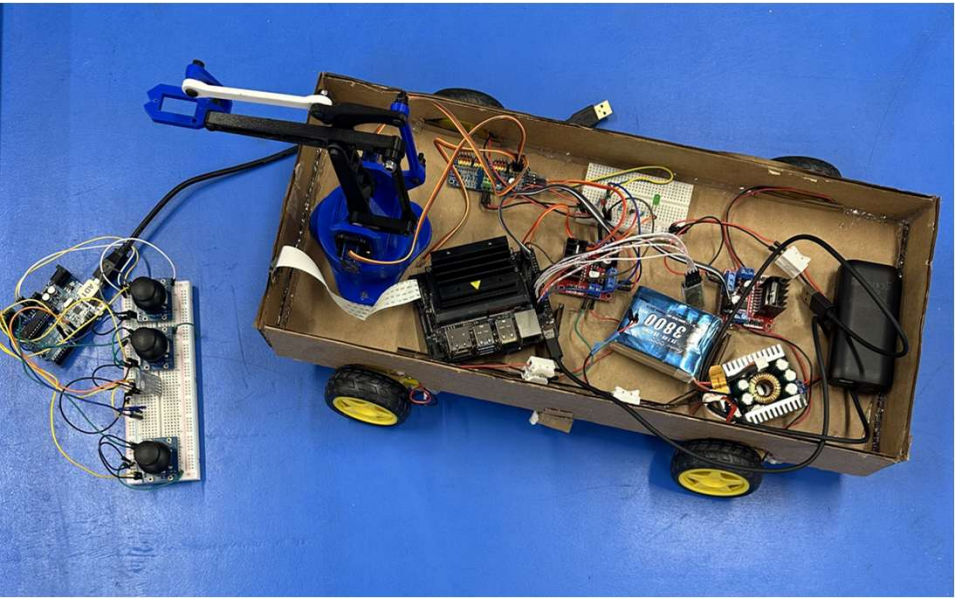
The Jetson Nano was better than the computer because at first the Jetson took 222 milliseconds to stream the video, then it went down about 160 milliseconds to stream. Whereas, the computer would constantly take about 220 milliseconds to 240 milliseconds. But the computer with the dedicated graphics card was actually competing against the Jetson Nano because both of them have dedicated graphics cards but the computer has better hardware to process the information faster. Raspberry Pi failed to handle the vehicle control commands as well as live streaming. Hence the Jetson is found to be the best option to carry out both the jobs without putting an actual computer onboard.

Revised Solution and Prototype/Model

I started off initially with the chassis having 3 pairs of wheels. Out of the 3 pairs, one would be at the very front and would be motorized. Then the pair in the middle would be just connected by an axle. Lastly the third pair would be connected to motors. But after test running it properly, I found out that the two wheels in the middle would cause too much drag and the motors could hardly move the rover right or left. Even the rover was failing to move forward - backward. So I removed the skewers they were connected to. Initial wheels setup was assuming that it would help in better weigh distribution and without them, the center would sag and would put a bit too much weight on the front and back pairs of wheels. So, in conclusion, I am using two pairs of wheels and all the wheels are motorized.

For the arm, I first thought of putting a servo at each joint or point of movement. But that resulted into putting too much weight on the servo on the base for forward and backward movement. So then I changed to a design that had all the servos on the base of the arm and there was no extra weight at the ends or at any join other than the ones at the base for all the movement.

For the boards, I at first thought of using an Arduino Mega 2560 Rev3 with a Raspberry Pi. But then I realized that Arduino was fine but the Raspberry Pi was very slow at video streaming and handling so many motor operation command flows. That is why, I used the Jetson Nano made by NVIDIA. It is basically a computer with a better GPU and is faster and better at wireless communication and live video streaming than most other developer boards. But one downside of using the Jetson Nano is that it does not have Analog Pins for the antennas I was about to use to connect with the Arduino. As the antennas did not work, I had to use bluetooth instead which now has a shorter range of only about 98 feet. During mid design, I had thought of using a bluetooth usb chip for RFCOMM based communication. But while trying this option, Jetson failed to build the Kernel required for RFCOMM. Hence, I switched to the HC05 bluetooth module although the chip had an increased range of about 40 feet.



The revised and final design:

- ❑ Vehicle with 4 motorized wheels
- ❑ Arm with all 3 servo motors at the base
- ❑ Adafruit board for servo control
- ❑ Jetson Nano for large command set for all 7 motors along with video streaming
- ❑ HC05 bluetooth modules for steady communication between control and rover modules
- ❑ 3 LEDs to show rover boot status and readiness to accept commands from control module
- ❑ 8V battery source to power motor board and an additional power source for Jetson

Discussion

First of all, initial 6-wheel design was revised with 4-wheel design to reduce the drag resulting into significant challenge with rover movement. The 2 middle wheels, which were non-motorized, were adding more drag. However, to have better mobility performance and work on rough terrains, the complete wheel structure can be modified with mecanum wheels connected by chain structure and with bigger surface area of each tire.

For the computer with shared graphics card, the range of time was from 223 milliseconds to 242 milliseconds. But the Jetson took even less time and had a range of streaming data in 152 milliseconds to 222 milliseconds. The Jetson is performing better because, it has a dedicated graphics card and it only does one activity at a time. The Raspberry Pi failed to process large amount of rover movement commands along with video streaming. Hence, Jetson was the best option considering the demands of this project. However, the computer with dedicated graphics card is even faster than the Jetson Nano. The timing range for that computer was from 117 milliseconds to 137 milliseconds. This helps us to conclude that the dedicated graphics card is the best for fast camera streaming and an actual computer is better than the NVIDIA Jetson Nano developer board for the real-life rovers.

Conclusion

My problem was that there was challenge for human in doing certain tasks like extra terrestrial tasks, very deep underwater tasks, and mine exploration tasks etc. frequency-based to help all the people who are risking their lives or to achieve extra terrestrial missions, I made a rover explorer that will explore for them.

With some modifications, it can go underwater and collect samples for research and with another different set of modifications, it can also do extra terrestrial exploration purposes. Some other things that I want to add to the project would be various sensors to measure the environmental factors for extra terrestrial and mine exploration purposes.

One other thing I would like to add to my project would be a lab where the samples can be tested in certain ways to make sure it is real and if it something new or not. This could be used in any of the mentioned cases.

I may also need to add suspension, so it can travel through uneven surfaces. To have better mobility performance and work on rough terrains, the complete wheel structure can be modified with mecanum wheels connected by chain structure and with bigger surface area of each tire.

And on scaling the thing up, I may need a Bluetooth module that is at least a 100 times better to do it for certain purposes or radio frequency-based communication for further enhancement. In conclusion, I can upgrade this model to make it near real life and turn it into a full-scale explorer to help with the mission statement in future.

Last but not the least, the rover module can be loaded with AI based object detection & identification for autonomous navigation as well as target sample identification for further analysis.

Reflection/Application

From doing this project, I learnt how to do planning. I learnt how to write complex codes with python, and some different ways of wireless communication. From the project I realized that the debugging of the codes might take longer than planned and I should have kept some planned buffer time to accommodate the same to avoid last minute rush.

Some of the next steps of research for my project are that (1) To have better mobility performance and work on rough terrains, the wheel structure can be modified with mecanum wheels with bigger surface area of each tire to allow strafing and on-spot rotation. (2) I am going to add a portable lab so that the tests from the samples can be done on spot for future analysis. (3) I am going to research about how to use AI with the Jetson Nano to allow the rover to know what the obstacle is, to know what samples to collect and what samples not to collect, etc. This will allow autonomous driving and sample collection for the rover because it can work with its own mind.

The rover can be used in many situations in everyday life. For e.g. (1) the rover can be used by firefighters to extract trapped human & things in a heavy fire zone and also the rover can be mounted with extinguisher to help contain the fire, (2) the rover can be sent to chemical or nuclear hazardous areas instead of human, (3) the rover can be sent to new or risky mines for exploration, (4) the rover can be used mounted with weapons and sent to war zone to fight enemy etc. In all these use cases of real & everyday life, the rover can perform human like activities and help saving human life.

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